

IEEE-CSR

A 5G-IoT Cyber Range Platform with Integrated Learning Modules

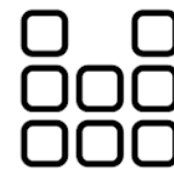
Stylianos Karagiannis, Sarantis Kalafatidis, Giannis Ledakis, Emmanouil Magkos, Christoforos Ntantogian



Dr. Stylianos Karagiannis (PhD)
Ionian University, Corfu, Greece



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NITRO 5G-IoT: The Next-Generation Cyber Range

Bridging theoretical 5G knowledge and practical cybersecurity skills through full-stack emulation and integrated learning.

A unified digital environment for modern network defense.



ISOMETRIC 5G NETWORK EMULATION CORE.

PROJECT PARTNERS



ΠΑΝΕΠΙΣΤΗΜΙΟ ΠΕΙΡΑΙΩΣ
UNIVERSITY OF PIRAEUS

LOCATION COUNTRY: Greece



The **Systems Security Laboratory (SSL)**, representing the University and its research, is an innovative security and systems analysis laboratory.

[Read more](#)



IONIAN
UNIVERSITY

LOCATION COUNTRY: Greece



This project partner will be responsible for the 5G cyber range deployment, representing and fulfilling the role of the 5G cyber range university for Ionian University.

[Read more](#)



HELLENIC
CYBERSECURITY
INSTITUTE (HCSI)

LOCATION COUNTRY: Greece



The **Hellenic Cybersecurity Institute** leads work in advancing cybersecurity practices for a more resilient digital society.

[Read more](#)

uni.systems

LOCATION COUNTRY:      

Greece (Headquarters) with Subsidiaries in:
Greece (HQ), Belgium, Luxembourg, Romania, Spain, Italy

An ICT solutions provider with expertise in constitution for global subsidiaries in Luxembourg, Spain, and other strong technology sectors.

[Read more](#)



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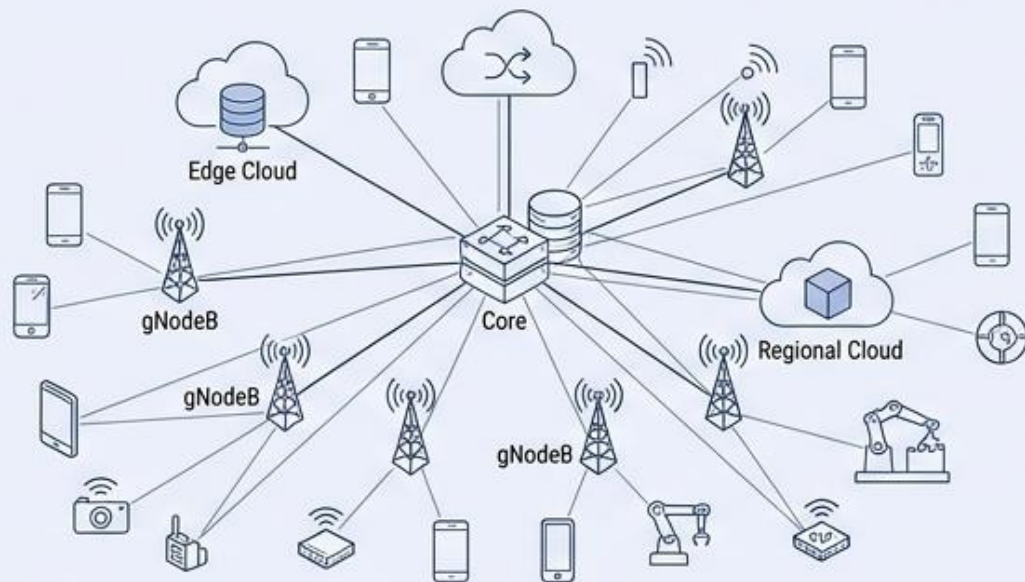
LOCATION COUNTRY:    

Greece (Headquarters) with Subsidiaries in:
Greece (HQ), Romania, Italy

A dynamic software integrator that provides technology, architecture, and support services, as well as a range of platform solutions.

[Read more](#)

The 5G Scale



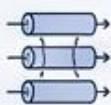
Service-Based Architecture (SBA)

Decoupled control plane functions communicating via RESTful APIs. Enhances flexibility and scalability.



Network Functions Virtualization (NFV/SDN)

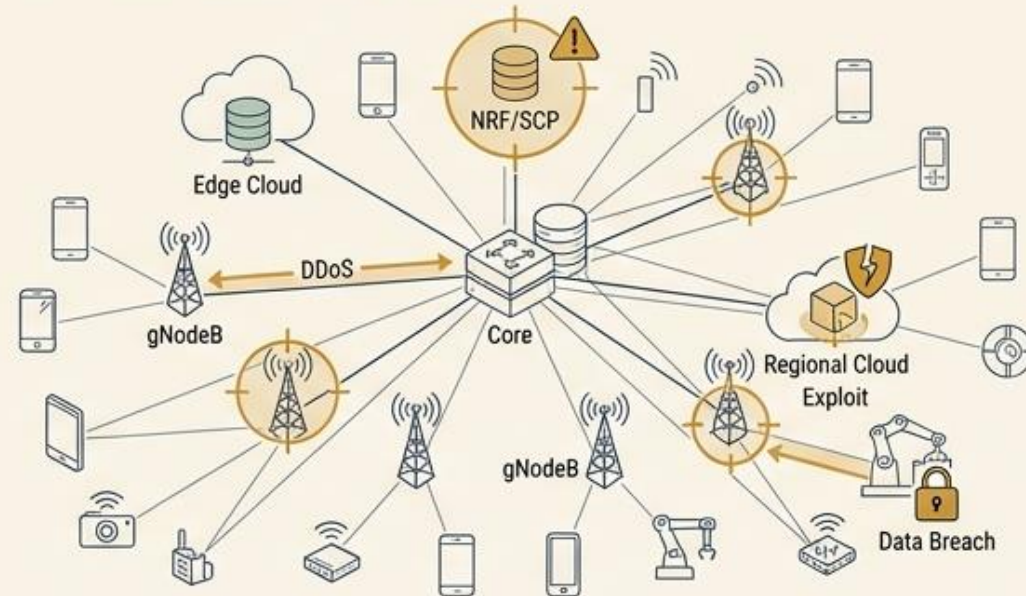
Virtualizes network appliances, separating software from hardware. Enables dynamic resource allocation.



End-to-End Network Slicing

Creates logically separated virtual networks on shared physical infrastructure for diverse use cases.

The Vulnerability Cost



Cross-slice information leakage

Unintended data flow between isolated network slices, compromising confidentiality and integrity.



NGAP/PCF signaling floods

Overwhelming control plane interfaces with excessive signaling traffic, causing denial of service.



Open RAN control-plane manipulation

Exploiting open interfaces to inject malicious configurations or disrupt radio resource management.



ML-IDS dataset poisoning

Injecting adversarial data into training sets to subvert intrusion detection systems and bypass security.

THE GAP: Educational platforms remain severely fragmented. We lack replicable laboratory platforms, practical full-stack workflows, and robust testbeds for adversarial Machine Learning.

The 5G Training Landscape

Platform	Full-Stack Emulation	Progressive Learning	Containerized Reproducibility	Adversarial AI Ready	Unified Web Interface
Cyber5Gym	✓	—	—	✗	✗
SPIDER	✓	✓	✗	✗	✗
Open 5G Testbed	✓	—	✓	✗	✗
NITRO 5G-IoT	✓	✓	✓	✓	✓

While others solve parts of the equation,
NITRO provides a holistic, unified ecosystem.





Replicable Ecosystem

Docker-based containerization ensures identical, repeatable experiments and strict user isolation.

NITRO Web Platform

Integrated 5G-IoT
Cyber Range & CTF
Environment



Full-Stack Emulation

Incorporates User Equipment (UE), Radio Access Network (RAN), and the 5G Core in a single environment.



Modular Progression

Fuses offensive and defensive scenarios with gamified performance scoring.

NITRO connects complex technical tasks directly to structured learning modules without requiring users to switch contexts or manage underlying infrastructure.

4. Orchestration Engine

Centralized Web Portal for VM deployment and lifecycle management.

3. 5G Core Network

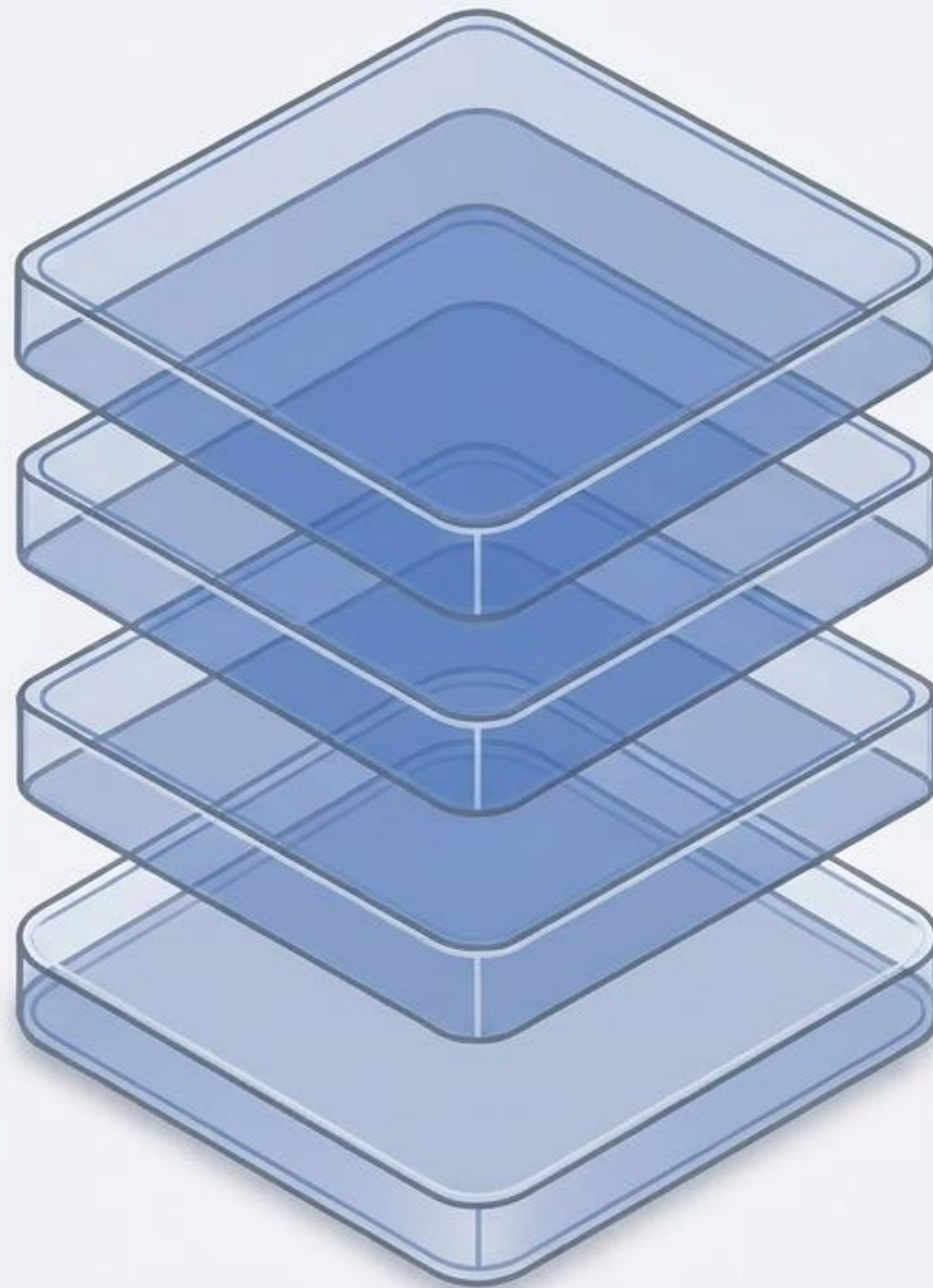
Powered by Open5GS.
Provides realistic service-based architecture and core network functions.

2. Access Network

Powered by UERANSIM. Delivers high-fidelity User Equipment (UE) and gNB emulation.

1. Infrastructure Foundation

Docker Containerization ensures user isolation and identical, repeatable exercise conditions.



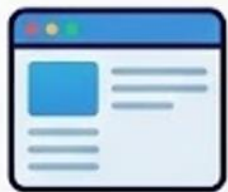
Full-stack emulation
ensures true
reproducibility,
moving beyond
theoretical threat
models to
high-fidelity
operational realities.

The Telemetry & Provisioning Loop



EXERCISE SYSTEM WORKFLOW

1. User Access & Configuration



Users Access Nitro Portal Web Interface

Participants use the portal to read manuals and configure their specific exercise environments.

2. Automated Testbed Provisioning



Virtual Machines Deploy in Testbed

The system automatically provisions VMs and establishes an SSH Proxy Server for secure connectivity.

3. Secure Remote Interaction



Users Connect via Secure SSH Tunnel

Remote access is mediated through the SSH Proxy to ensure controlled interaction with the exercise VMs.

4. Real-Time Telemetry Collection



Security Agents Capture Environment Data

Agents in each VM feed security logs and telemetry directly to the Security Manager for indexing.

5. Analysis & Performance Scoring



Gamification System Calculates User Scores

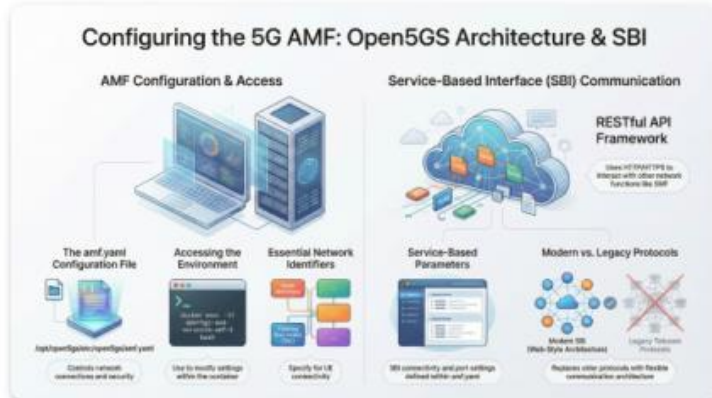
Indexed security data is evaluated to provide real-time visibility and final performance metrics.



Demonstration

Exercise: 03. Execute an Initial Ping to a Public IP Address

Exercise Preview



Click to enlarge

Information

Status
Started

Start Date
3/3/2026

Difficulty
Beginner

Points
5

Module
Module 01 - Introduction to 5GCore and User Equipment (UE)

Instructions

Complete the tasks in the terminal and submit your answer below.

Step 3: Execute an Initial Ping to a Public IP Address

Before analyzing complex protocol behavior, we first confirm basic connectivity. The ping command is one of the simplest tools in networking, but in 5G it hides significant complexity under the surface.

Action: Execute an Initial Ping to a Public IP Address

Command:

```
ping 8.8.8.8
```

and in another terminal in the same machine run `tshark -i any icmp`

Expected Output: Ping responses showing round-trip times

(e.g., 64 bytes from 8.8.8.8: icmp_seq=1 ttl=117)

MATERIALS



Demonstration

Virtual Machines Management

Virtual Machine limit: 1 / 2 [Help](#) [+ Create a Virtual Machine](#)

IonianUni-5G
CPU: 2 | Memory: 4096MB

Connect to Virtual Machines

VM Status: IonianUni-5G RUNNING [Stop VM](#) [Refresh](#)

Sudo password:

Terminal 1

Select VM for Terminal 1

IonianUni-5G

[Connect](#)

Terminal 2

Select VM for Terminal 2

IonianUni-5G

[Connect](#)

Active SSH Sessions

VM Name	Status	Duration	Action
IonianUni-5G	Connected	00:00:02	Disconnect
IonianUni-5G	Connected	00:00:00	Disconnect

Terminal 1 - IonianUni-5G

```

                                NAMES
Safcd1b49f10  gradient/ueransim:3.2.6    "/entrypoint.sh ue -..." Up 4 minutes
                                open5gs-and-ueransim-ues1-1
ef8098a8d1ea  gradient/ueransim:3.2.6    "/entrypoint.sh gnb"      Up 4 minutes
                                gnb1
18bb99554c82  gradient/open5gs-webui:2.7.5 "node server/index.js"    Up 5 minutes  0.0.0.0
:9999->9999/tcp, [::]:9999->9999/tcp open5gs-and-ueransim-webui-1
f3d85f46d170  mongo:4.4                  "docker-entrypoint.s..." Up 5 minutes  27017/t
cp                                open5gs-and-ueransim-mongo-1
```

Terminal 2 - IonianUni-5G

Your Answer

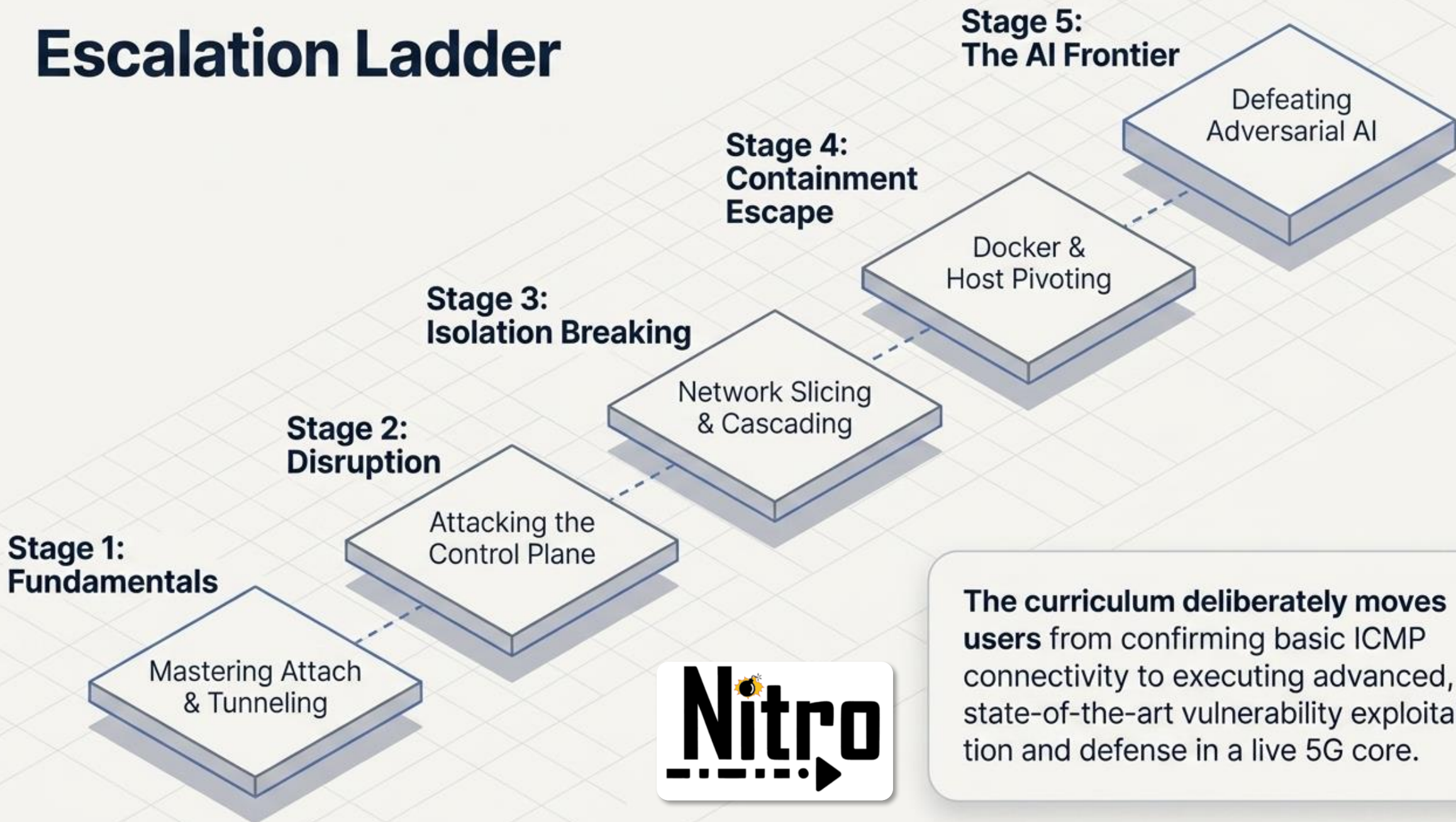
Enter your answer

[Back](#) [Submit](#)

Enter to submit
Shift+Enter for new line

[MATERIALS](#)

Escalation Ladder

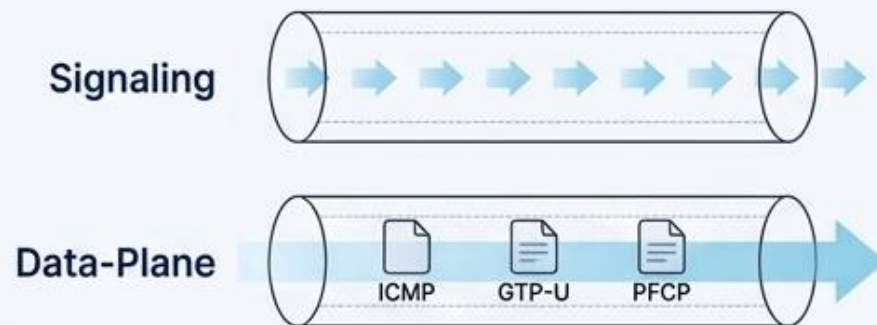


Stage 1: Attach & Tunneling

Fundamentals

The Action: Users send ICMP packets over virtual tunnels, actively capturing and inspecting GTP-U and PFCP traffic.

The Insight: By comparing signaling versus data-plane behavior, encapsulation and interface-specific QoS become tangible.



Stage 2: Disrupting the Control Plane

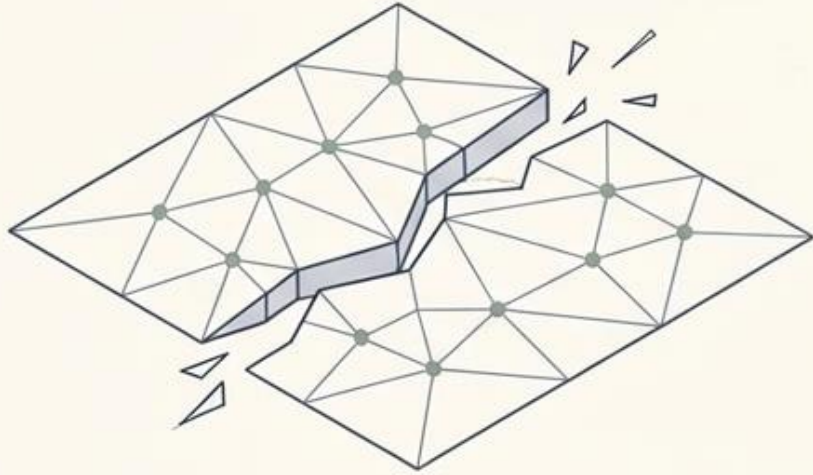
Resource Exhaustion

The Action: Users execute parallel UE registration scripts and **5G replays** to perform DDoS attacks on the AMF.

The Insight: Observing resource exhaustion and SCTP rejections teaches the direct operational impact of signaling floods and reinforces mitigation strategies.



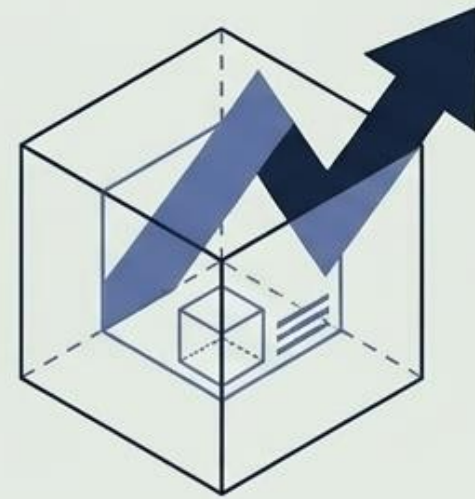
Stage 3: Breaking Network Slices



The Action: Modifying configurations across the AMF, SMF, and UPF to measure bandwidth enforcement and attempt session hijacking.

The Insight: Demonstrates how minor misconfigurations create high-impact cascading vulnerabilities and cross-slice information leakage.

Stage 4: Escaping Containerized NFs



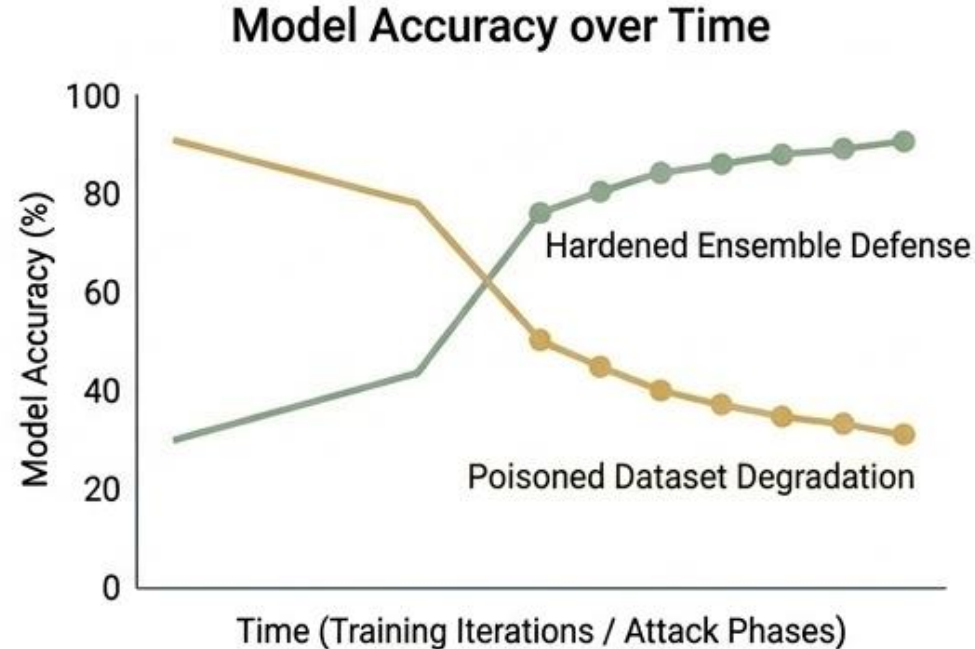
The Action: Performing a mount/chroot sequence to pivot from a restricted Docker container directly to host root.

The Insight: A definitive, interactive demonstration proving why least-privilege deployment and capability restrictions are mandatory in 5G NFV environments.

Stage 5: The AI Frontier

The Attack

Users poison datasets and perform inference/time evasion maneuvers to degrade ML-based Intrusion Detection Systems analyzing 5G NetFlow data.



The Defense

Users actively retrain the models using perturbed data and ensemble classifiers to harden the system.

Synthesis Insight

Comparing evasion rates before and after hardening provides an empirical baseline for practical ML resilience in 5G networks—a major capability missing from existing cyber ranges.

Synthesis Insight

TABLE I
SURVEY QUESTION SUMMARY STATISTICS (7-POINT LIKERT SCALE)

Question	Category	Mean	Median	SD	%>5
A1. Overall satisfaction with the cyber range	Course content and structure	6.17	7.00	1.13	75.0
A2. Quality of instructional materials		6.42	7.00	0.78	91.7
A3. Clarity of instructional materials		6.33	7.00	0.96	75.0
A4. Practical relevance of the content		6.43	7.00	0.66	91.3
A5. Alignment of course design and content with objectives		6.50	7.00	1.29	91.7
B1. Accessibility of the learning platform	Learning Platform	6.25	7.00	1.42	91.7
B2. Ease of navigation of the learning platform		6.46	7.00	1.32	87.5
B3. Performance and reliability of the platform		5.96	6.00	1.55	83.3
B4. Visual appeal of the platform		6.48	7.00	1.04	91.3
B5. Interactivity & engagement opportunities on the platform		6.25	7.00	1.42	87.5
B6. Interaction facilitated between learners		6.58	7.00	0.78	91.7
C1. Transparency of the examination process	Evaluation & Recognition	6.58	7.00	0.72	95.8
C2. Fairness of the examination process		6.21	7.00	1.35	87.5
C3. Value of the (attendance) certificate		5.42	6.00	1.72	58.3
D1. Relevance of skills and knowledge to job role	Impact	6.00	6.00	1.14	75.0
D2. Extent course enhanced knowledge and skills		6.08	6.00	0.93	70.8
D3. Likelihood to further explore the topic		6.38	7.00	0.88	83.3
D4. Likelihood to recommend to others		6.58	7.00	0.72	95.8

Performance & Impact Dashboard (N=24)



of users scored Above Average or High. Zero low scores recorded.

95.8%

Praised the transparency and fairness of the evaluation process.

91.7%

Highlighted top-tier quality of materials, platform accessibility, and strict alignment with course objectives.

THE CERTIFICATE INSIGHT: 58.3% placed high value on the attendance certificate itself. The intrinsic value of the hands-on skills learned far outweighed the extrinsic value of the paper credential.

Introduction: The 5G Talent Gap

Sobering Reality

Networks are Virtualized, but Training is Not

Dangerously Fragmented Skills

Missing Real-World Replications

Isolated Labs & Standalone Datasets Fail



Urgent Need & The NITRO Solution

Paradigm Shift: Beyond Theory

Bridging Classroom to Reality

Replicable Laboratory Ecosystem



UNIFIED 5G CYBER DEFENSE

The NITRO 5G-IoT platform bridges the chasm between theoretical knowledge and the realities of live 5G core security.

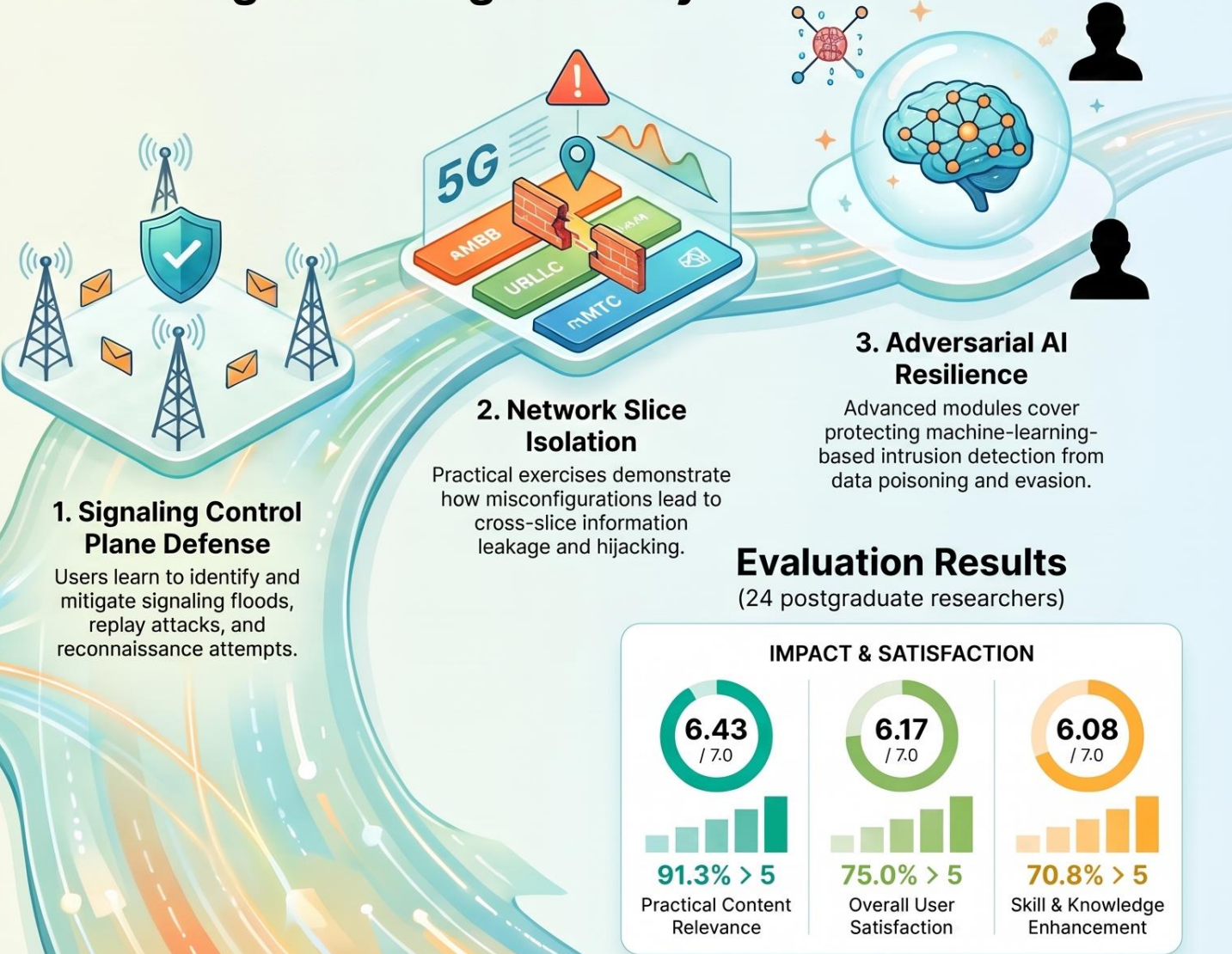
Nitro

NITRO 5G-IoT: Bridging the Gap in 5G Cybersecurity Training

The 3-Step Training Workflow



The 3-Stage Learning Pathway



IEE-CSR

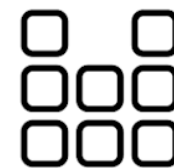
Thank you for your Attention!



Dr. Stylianos Karagiannis (PhD)
Ionian University, Corfu, Greece



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